



IT-SIMPLERA INSTITUTE
Red Team Cyber Security Internship Program

WEEK 04 ASSIGNMENT REPORT
Reconnaissance & OSINT Attack Surface Mapping

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Introduction

This report documents the Week 04 assignment of the Cyber Security Analyst Internship Program at IT-Simplera Institute, supervised by Talal Rehan. Building on the reconnaissance and scanning fundamentals covered in Week 03, this assignment focuses on applying passive and active reconnaissance techniques, along with Open Source Intelligence (OSINT) methods, against an authorized practice target.

The purpose of this assignment is to simulate the first phase of a real-world penetration testing or bug bounty engagement: gathering publicly available information about a target organization, mapping its attack surface, and identifying potential security risks all without performing any exploitation. The work is organized into ten tasks covering WHOIS/DNS enumeration, Google Dork investigation, subdomain discovery, live host verification, screenshot collection, technology fingerprinting, attack surface diagramming, risk identification, and reconnaissance automation.

Task 1 — Passive Reconnaissance

Passive reconnaissance gathers information about a target without directly interacting with its systems, reducing the chance of detection. In this task, publicly available sources are used to profile the chosen target's organization, domain, hosting, and security posture.

1.1 Organization Overview

HTB (Hack The Box): online platform, offers vulnerable VMs and labs for legal hands on pentest practice. Has free and paid tiers, includes Academy for guided learning plus live CTF style boxes for real recon and exploitation skill building. Widely used and respected in cybersecurity hiring, good authorized target for my recon assignment.

1.2 Domain, IP Ownership & Registrar

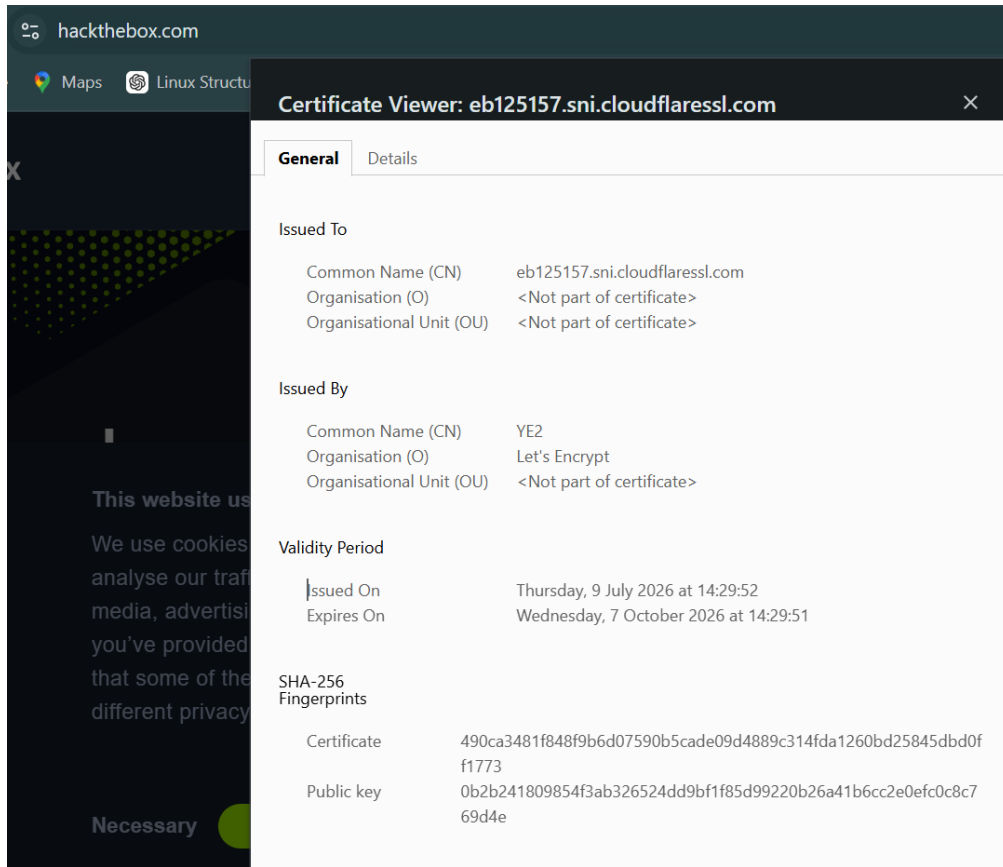
Attribute	Value
Main Domain	<i>www.hackthebox.com</i>
IP Address	<i>109.176.239.69, 109.176.239.70</i>
ASN (if available)	<i>From Shodan / bgp.he.net</i>
Registrar	<i>Amazon Registrar, Inc. (http://registrar.amazon.com)</i>
Hosting Provider	<i>Cloudflare (nameservers point to cloudflare, so most likely it is proxied behind it)</i>
Nameservers	<i>cody.ns.cloudflare.com, jill.ns.cloudflare.com</i>

```
(kali㉿kali)-[~]
$ whois hackthebox.com
Domain Name: HACKTHEBOX.COM
Registry Domain ID: 1589213536_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.registrar.amazon
Registrar URL: http://registrar.amazon.com
Updated Date: 2026-05-15T15:03:57Z
Creation Date: 2010-03-18T03:26:29Z
Registry Expiry Date: 2027-03-18T03:26:29Z
Registrar: Amazon Registrar, Inc.
Registrar IANA ID: 468
Registrar Abuse Contact Email: trustandsafety@support.aws.com
Registrar Abuse Contact Phone: +1.2024422253
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Name Server: CODY.NS.CLOUDFLARE.COM
Name Server: JILL.NS.CLOUDFLARE.COM
```

```
(kali㉿kali)-[~]
$ nslookup hackthebox.com
Server:      192.168.18.1
Address:     192.168.18.1#53

Non-authoritative answer:
Name:   hackthebox.com
Address: 109.176.239.69
Name:   hackthebox.com
Address: 109.176.239.70
```

1.3 SSL Certificate Information



The screenshot shows a web browser window with the address bar displaying "hackthebox.com". A "Certificate Viewer" window is open, showing details for the certificate "eb125157.sni.cloudflaressl.com". The viewer has two tabs: "General" (selected) and "Details".

General

Issued To

Common Name (CN)	eb125157.sni.cloudflaressl.com
Organisation (O)	<Not part of certificate>
Organisational Unit (OU)	<Not part of certificate>

Issued By

Common Name (CN)	YE2
Organisation (O)	Let's Encrypt
Organisational Unit (OU)	<Not part of certificate>

Validity Period

Issued On	Thursday, 9 July 2026 at 14:29:52
Expires On	Wednesday, 7 October 2026 at 14:29:51

SHA-256 Fingerprints

Certificate	490ca3481f848f9b6d07590b5cade09d4889c314fda1260bd25845dbd0f1773
Public key	0b2b241809854f3ab326524dd9bf1f85d99220b26a41b6cc2e0efc0c8c769d4e

1.4 Technologies Identified

The screenshot displays the Wappalyzer extension interface on the website hackthebox.com. The interface is divided into two main sections: a left panel showing the website's content and a right panel listing identified technologies.

Left Panel (Website Content):

- Header: THEBOX, Why HTB, Platform
- Section: d and scale a cvbe
- Text: This website uses cookies. We use cookies to personalise content and ads, to analyse our traffic. We also share information about media, advertising and analytics partners who may use the information you've provided to them or that they've collected from their site, including that some of the cookies may transfer information to different privacy standards.

Right Panel (Technologies Identified):

- TECHNOLOGIES** (Selected) | MORE INFO | Export
- CMS**
 - HubSpot CMS Hub
- Analytics**
 - LinkedIn Insight Tag
 - HockeyStack
 - Google Analytics
 - Facebook Pixel
- JavaScript frameworks**
 - GSAP
- Advertising**
 - Twitter Ads
 - AdRoll
- Tag managers**
 - Google Tag Manager
- Live chat**
 - HubSpot Chat
- JavaScript libraries**
 - Goober
- Issue trackers**
 - Sentry
- Font scripts**
 - Adobe Fonts
- Miscellaneous**
 - Open Graph
 - HTTP/3
- CDN**
 - Cloudflare
- Marketing automation**
 - HubSpot
- jQuery** 1.7.1
- core-js** 3.32.2
- Cookie compliance**
 - Cybot
 - Cookiebot 1
 - Cookie Control
 - AdRoll CMP System
- Appointment scheduling**
 - Chili Piper
- Retargeting**
 - AdRoll
- Performance**
 - Priority Hints

1.5 Email Security Records (SPF, DKIM, DMARC)

Record	Present? (Y/N)	Value / Notes
SPF	Y	<i>v=spf1 include:_u.hackthebox.com._spf.smart.ondmarc.com ~all</i>
DKIM	Y	google._domainkey.hackthebox.com TXT "v=DKIM1; k=rsa; p=MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAgTAAoBzIPD1UHG6mS3VJxw4HnfVXgLDwkID3ugGo0NeSkWD+G1TJGtS0ekMcG9sNNZ9aiWnXnUITHBiHlznGdHOT99d8+hFpfsdx3GFfeFTyPJTxlzjUd4qzYgZZ9CSA86/1GCQk+2m5xqTRc+phWQXfnxwbXdXRbjpE6dhDxCzgmDagQc3y7KU1FQvbLuXHs5H3A7vzV0AIOLcgOxk5A8nQlvHWA+ZPhnHTLfPlir0ByxCX9UKvDrQsDLQU98afejwxEjFPZsl5xqlqEobaf7zy5ul0rC0k0kuZf3LmOsQ/rIjVn4HqAxmxuk/QwCCgA5aMYKI3I7h9XPbQjxIwIDAQAB"
DMARC	Y	<i>v=DMARC1; p=quarantine; pct=100; sp=quarantine; rua=mailto:69d7f41b@inbox.ondmarc.com; ruf=mailto:69d7f41b@inbox.ondmarc.com; adkim=r; aspf=r; fo=1; rf=afrf; ri=3600</i>

```
;; ANSWER SECTION:
hackthebox.com.      300      IN       TXT      "brevo-code:293be74b395c496ce75d12a22ce91aee"
hackthebox.com.      300      IN       TXT      "v=spf1 include:_u.hackthebox.com._spf.smart.ondmarc.com ~all"
```

```
;; ANSWER SECTION:
google._domainkey.hackthebox.com. 2790 IN TXT      "v=DKIM1; k=rsa; p=MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAgTAAoBzIPD1UHG6mS3VJxw4HnfVXgLDwkID3ugGo0NeSkWD+G1TJGtS0ekMcG9sNNZ9aiWnXnUITHBiHlznGdHOT99d8+hFpfsdx3GFfeFTyPJTxlzjUd4qzYgZZ9CSA86/1GCQk+2m5xqTRc+phWQXfnxwbXdXRbjpE6dhDxCzgmDagQc3y7KU1FQvbL" "uXHs5H3A7vzV0AIOLcgOxk5A8nQlvHWA+ZPhnHTLfPlir0ByxCX9UKvDrQsDLQU98afejwxEjFPZsl5xqlqEobaf7zy5ul0rC0k0kuZf3LmOsQ/rIjVn4HqAxmxuk/QwCCgA5aMYKI3I7h9XPbQjxIwIDAQAB"
```

```
;; ANSWER SECTION:
_dmarc.hackthebox.com. 281      IN       CNAME     _dmarc.hackthebox.com._dmarc.smart.ondmarc.com.
_dmarc.hackthebox.com._dmarc.smart.ondmarc.com. 1940 IN TXT      "v=DMARC1; p=quarantine; pct=100; sp=quarantine; rua=mailto:69d7f41b@inbox.ondmarc.com; ruf=mailto:69d7f41b@inbox.ondmarc.com; adkim=r; aspf=r; fo=1; rf=afrf; ri=3600"
```

Task 2 — Advanced WHOIS & DNS Enumeration

This task expands on WHOIS lookups and performs full DNS record enumeration to understand the target's mail infrastructure, name resolution, and email authentication configuration.

2.1 WHOIS Summary

Field	Value
Registrar	Amazon Registrar, Inc. (http://registrar.amazon.com)
Domain Creation Date	2010-03-18
Domain Expiration Date	2027-03-18

```
(kali㉿kali)-[~]  
$ whois hackthebox.com  
Domain Name: HACKTHEBOX.COM  
Registry Domain ID: 1589213536_DOMAIN_COM-VRSN  
Registrar WHOIS Server: whois.registrar.amazon  
Registrar URL: http://registrar.amazon.com  
Updated Date: 2026-05-15T15:03:57Z  
Creation Date: 2010-03-18T03:26:29Z  
Registry Expiry Date: 2027-03-18T03:26:29Z  
Registrar: Amazon Registrar, Inc.  
Registrar IANA ID: 468
```

2.2 DNS Record Enumeration Table

Record Type	Value(s)	Notes
A	109.176.239.70, 109.176.239.69	IPv4 addresses (2 A records)
AAAA	No AAAA records found	Domain does not have an IPv6 address
MX	1 aspmx.l.google.com. 5 alt1.aspmx.l.google.com. 5 alt2.aspmx.l.google.com. 10 alt3.aspmx.l.google.com. 10 alt4.aspmx.l.google.com.	Google Workspace mail servers
NS	jill.ns.cloudflare.com cody.ns.cloudflare.com	Authoritative name servers
TXT	v=spf1 include:u.hackthebox.com.spf.smart.ondmarc.com ~all (plus multiple verification records for Google, Slack, Atlassian, Apple, Facebook, 1Password, etc.)	SPF and service verification records
CNAME	No record found	Root domain usually has no CNAME
SOA	cody.ns.cloudflare.com. sysops.hackthebox.eu. 2410117829 10000 2400 604800 1800	Start of authority record

```
;; ANSWER SECTION:
hackthebox.com.      300      IN       A       109.176.239.70
hackthebox.com.      300      IN       A       109.176.239.69
```

```
;; QUESTION SECTION:
;hackthebox.com.          IN       AAAA

;; AUTHORITY SECTION:
hackthebox.com.      1013     IN       SOA      cody.ns.cloudflare.com. sysops.hackthebox.eu. 2410117829 10000 2400 604800 1800
```

```
;; ANSWER SECTION:
hackthebox.com.      21600    IN       NS       jill.ns.cloudflare.com.
hackthebox.com.      21600    IN       NS       cody.ns.cloudflare.com.
```

```
;; ANSWER SECTION:
hackthebox.com.      300      IN       TXT      "figma-domain-verification=d1cdada4c1c15f86d6b7cc65fbd95c5e34502a4c8bb6950368a680433f521c02-1763047074"
hackthebox.com.      300      IN       TXT      "atlassian-sending-domain-verification=469d03af-e04b-4609-a67a-0c4161077605"
hackthebox.com.      300      IN       TXT      "atlassian-domain-verification=BOEsvQIaiNXegte2TKsFBSjaCETV5f2F7rcvTiirvI/v9l56nU97DGOQfJIXWJJN"
hackthebox.com.      300      IN       TXT      "slack-domain-verification=XJrLipfXUDw7kCGV5rK68GS1d92cVqE51H4UZ6fP"
hackthebox.com.      300      IN       TXT      "google-site-verification=p8Ke_ETLIL22PW1uxiN0fE_-hLZ2LQd9KmdQs9bARa8"
hackthebox.com.      300      IN       TXT      "apple-domain-verification=nJdetwYEnUhgLT8W"
hackthebox.com.      300      IN       TXT      "google-site-verification=VdT_w4v3nriJ_MIknzhw01nVw10T_DFVf3JJP8yVD1C8"
hackthebox.com.      300      IN       TXT      "brevo-code:293be74b395c496ce75d12a22ce91aee"
hackthebox.com.      300      IN       TXT      "google-site-verification=5Zgk1p8QogX_Q_WaZ3PNUx0jaBMcR2CXXD1SYeHk_uM"
hackthebox.com.      300      IN       TXT      "oneuptime-verification=gKfEKTDuakVmbIHNrkpU"
hackthebox.com.      300      IN       TXT      "facebook-domain-verification=2ic10ka7c5jbl7ou7nva60919jn15w"
hackthebox.com.      300      IN       TXT      "1password-site-verification=4ZZGMSMKHBCCTHWRCUW5TEFLCA"
hackthebox.com.      300      IN       TXT      "v=spf1 include:_u.hackthebox.com._spf.smart.ondmarc.com ~all"
```

```
;; AUTHORITY SECTION:
hackthebox.com.      657      IN       SOA      cody.ns.cloudflare.com. sysops.hackthebox.eu. 2410117829 10000 2400 604800 1800
```

2.3 Mail Provider & Email Security Validation

Task	Answer	Evidence
Identify the mail provider	Google Workspace (Gmail)	<i>The MX records point to aspmx.l.google.com, alt1.aspmx.l.google.com, alt2.aspmx.l.google.com, alt3.aspmx.l.google.com, and alt4.aspmx.l.google.com, which are Google's mail servers.</i>
Determine whether email security records are configured correctly	Yes, they are configured correctly	SPF: Present (v=spf1 include:_u.hackthebox.com._spf.smart.ondmarc.com ~all) DKIM: Present (valid selectors google._domainkey.hackthebox.com and mail._domainkey.hackthebox.com) DMARC: Present (v=DMARC1; p=quarantine; ...)

Task 3 — Google Dork Investigation

Google Dorking uses advanced search operators to uncover publicly indexed but potentially sensitive files, portals, and configuration data belonging to the target. At least 20 unique dorks must be tested and documented below.

3.1 Google Dork Results Table (minimum 20 entries)

#	Search Query (Dork)	Purpose	Result
1	<i>site:owasp.org filetype:pdf</i>	<i>Finds .pdf files</i>	<i>https://owasp.org/assets/financial-audits/OWASP_Audit_Report_2022.pdf</i>
2	<i>site:owasp.org filetype:docx</i>	<i>Finds .docx files</i>	<i>https://board.owasp.org/attachments/202306-finance-narrative.docx</i>
3	<i>site:owasp.org filetype:xlsx</i>	<i>Finds .xlsx(excel) files</i>	<i>https://board.owasp.org/attachments/202406-uncat-items.xlsx</i>
4	<i>site:uetmardan.edu.pk login</i>	<i>Finds login pages</i>	<i>https://cms.uetmardan.edu.pk/</i>
5	<i>site:portswigger.net filetype:pdf</i>	<i>Finds .pdf files</i>	<i>https://portswigger.net/kb/papers/huntingasynchronousvulnerabilities.pdf</i>
6	<i>site:portswigger.net inurl:api</i>	<i>Finds API pages</i>	<i>https://portswigger.net/web-security/api-testing</i>
7	<i>site:developer.mozilla.org swagger</i>	<i>Searches swagger docs</i>	<i>https://developer.mozilla.org/en-US/docs/Web/HTTP/Reference/Status/415</i>
8	<i>site:developer.mozilla.org openapi</i>	<i>Search openAPI references</i>	<i>https://developer.mozilla.org/en-US/docs/Glossary/API</i>
9	<i>site:kali.org filetype:pdf</i>	<i>Finds kali PDFs</i>	<i>https://bugs.kali.org/file_download.php?file_id=2829&type=bug</i>
10	<i>site:ubuntu.com filetype:pdf</i>	<i>Finds Ubuntu PDFs</i>	<i>https://ubuntu.com/legal/ubuntu-pro-description</i>
11	<i>site:docs.docker.com filetype:yaml</i>	<i>Finds .yaml files</i>	<i>No results found</i>
12	<i>site:docs.github.com api</i>	<i>Finds API documentation</i>	<i>https://docs.github.com/en/rest</i>

#	Search Query (Dork)	Purpose	Result
13	<i>site:nginx.org filetype:conf</i>	<i>Search configuration examples</i>	<i>https://trac.nginx.org/nginx/attachment/ticket/565/nginx.conf</i>
14	<i>site:apache.org filetype:pdf</i>	<i>Finds Apache PDFs</i>	<i>https://s.apache.org/muamq</i>
15	<i>site:python.org filetype:pdf</i>	<i>Finds python PDFs</i>	<i>https://www.python.org/m/files/2014_redacted.pdf/</i>
16	<i>site:nodejs.org filetype:json</i>	<i>Finds JSON files</i>	<i>No results found</i>
17	<i>site:hackthebox.com intitle:"login"</i>	<i>Search pages whose title contains "login".</i>	<i>https://academy.hackthebox.com/course/preview/login-brute-forcing</i>
18	<i>site:hackthebox.com inurl:admin</i>	<i>Find URLs containing admin</i>	<i>No results found</i>
19	<i>site:hackthebox.com intext:"API"</i>	<i>Find pages that contain the text API.</i>	<i>https://www.hackthebox.com/blog/intro-to-mobile-pentesting</i>
20	<i>site:hackthebox.com intitle:"index of"</i>	<i>Look for open directory listings.</i>	<i>https://forum.hackthebox.com/t/index-of-file/3535/4</i>

Google

AI Mode **All** Images Videos Forums News Short videos More ▾ Tools

 Hack The Box :: Forums
<https://forum.hackthebox.com> › Tutorials › Other

index of file - #4 by addedie - Other

02-Dec-2020 — how i can get index of file? and what is it.

4.1 Tool Output Summary

Tool	Site	Subdomains found
Subfinder	uop.edu.pk	36
I used recon-ng instead of amass because amass was not working with me properly.	uop.edu.pk	27
Assetfinder	uop.edu.pk	28
Merged & Deduplicated Total	<i>≥ 30 required</i>	37

```
(kali㉿kali)-[~]
$ subfinder -d uop.edu.pk > uopDomains.txt
```

README Contributing License Security
Subfinder v2.14.0 (latest)
Enumerating subdomains for projectdiscovery.io

```
[INF] Current subfinder version v2.14.0 (latest)
[INF] Loading provider config from /home/kali/.config/subfinder/provider-config.yaml
[INF] Enumerating subdomains for projectdiscovery.io
[INF] Found 36 subdomains for projectdiscovery.io in 31 seconds 571 milliseconds
```

```
[recon-ng][practice][hackertarget] > run
```

UOP.EDU.PK

SUMMARY

```
[*] 27 total (27 new) hosts found.
```

```
(kali㉿kali)-[~]  
$ assetfinder --subs-only uop.edu.pk >> ~/uopDomains.txt
```

```
(kali㉿kali)-[~]
$ sort -u ~/uopDomains.txt -o ~/uopDomains.txt

(kali㉿kali)-[~]
$ wc -l ~/uopDomains.txt
37 /home/kali/uopDomains.txt
```

4.2 Subdomain Categorization

Category	Subdomains	Security Relevance
API	<i>None Identified</i>	<i>APIs may expose sensitive endpoints, authentication tokens, and confidential data if not secured.</i>
Development	<i>demo.uop.edu.pk</i>	<i>Development systems may contain debug features, test accounts, or unfinished code that attackers can exploit.</i>
Staging	<i>None Identified</i>	<i>Staging environments often mirror production and may accidentally expose sensitive information.</i>
Admin	<i>portal.uop.edu.pk, www.portal.uop.edu.pk, ums.uop.edu.pk, inventory.uop.edu.pk, admissions.uop.edu.pk</i>	<i>Administrative portals contain sensitive data and require strong authentication and access controls.</i>
VPN	<i>None Identified</i>	<i>VPN gateways provide remote access to internal resources and must be protected with strong authentication.</i>
Mail	<i>mail.uop.edu.pk</i>	<i>Mail servers should be protected with SPF, DKIM, and DMARC to prevent phishing and email spoofing.</i>
CDN	<i>None Identified</i>	<i>CDN services distribute content globally. Misconfiguration may expose cached content or infrastructure details.</i>
DNS	<i>ns1.uop.edu.pk, ns2.uop.edu.pk</i>	<i>DNS servers are critical infrastructure. Misconfigurations can lead to DNS spoofing or information disclosure.</i>

Task 5 — Live Host Verification

Not every discovered subdomain is active. httpx probes each subdomain over HTTP/HTTPS to confirm which hosts are live and captures key metadata for each, forming the basis for further investigation.

5.1 Live Host Summary Table

Subdomain	Status Code	Title	Web Server	Redirect	HTTPS	Tech Fingerprint
<i>etc.uop.edu.pk</i>	502	502 Bad Gateway	nginx/1.24.0	No	Yes	Nginx 1.24.0, Ubuntu
<i>portal.uop.edu.pk</i>	200	PAKTECH Complete ERP Solution	Microsoft-IIS/10.0	No	Yes	ASP.NET, IIS 10.0, Bootstrap, jQuery
<i>demo.uop.edu.pk</i>	200	University of Peshawar – Leading Public University in Pakistan	nginx/1.24.0	No	Yes	Next.js, React, Node.js, Bootstrap
<i>learn.uop.edu.pk</i>	303 → 200	Dashboard (Guest) Learning Management System	nginx/1.24.0	https://learn.uop.edu.pk/my/	Yes	Moodle, PHP, RequireJS
<i>baragali.uop.edu.pk</i>	200	Baragali Summer Camp - UoP	nginx/1.24.0	No	Yes	Next.js, React, Node.js
<i>szic.uop.edu.pk</i>	302 → 200	Account Suspended	LiteSpeed	https://szic.uop.edu.pk/cgi-sys/suspendedpage.cgi	Yes	LiteSpeed, HTTP/3
<i>mail.uop.edu.pk</i>	404	Error 404 (Not Found)	ghs	No	No	Google Hosted Services (GHS)
<i>ncepc.uop.edu.pk</i>	200	Home	Apache/2.4.54	No	No	Joomla, PHP 8.0.25, OpenSSL
<i>mosaic.uop.edu.pk</i>	200	<i>mosaic.uop.edu.pk</i> – Journal,	LiteSpeed	No	Yes	WordPress,

		Newsletter, IR				Elementor, PHP 8.2
isc.uop.edu.pk	200	International Studies, Baragali	LiteSpeed	No	Yes	WordPress , LiteSpeed
ums.uop.edu.pk	200	Home - UMS	LiteSpeed	No	Yes	WordPress , Elementor, Yeast SEO
bic.uop.edu.pk	200	BIC – University of Peshawar	LiteSpeed	No	Yes	WordPress , WooComm erce, Elementor
politicalscience.uop .edu.pk	200	Home - Politics	LiteSpeed	No	Yes	WordPress , Elementor
cpeccsc.uop.edu.pk	200	CPEC CSC Conference 2019	Apache	No	No	Apache, Bootstrap
khayaban.uop.edu. pk	200	(No title)	Apache	No	No	Apache HTTP Server
mrl.uop.edu.pk	403	Apache HTTP Server Test Page	Apache	No	No	Apache HTTP Server
ojs.uop.edu.pk	N/A	journals	Apache/2. 4.6	No	No	Open Journal Systems, PHP 7.3
mrc.uop.edu.pk	200	M R C – Migration Research Cell	LiteSpeed	No	Yes	WordPress , Elementor
ipcs.uop.edu.pk	200	IPCS UOP	Apache	No	No	Apache, Bootstrap, PHP 5.6
che.uop.edu.pk	200	College of Home Economics	Apache	No	No	Apache, Bootstrap, PHP 5.6
nceg.uop.edu.pk	200	National Centre of Excellence in Geology	Apache	No	No	Apache, DreamWe aver
journals.uop.edu.p k	200	Journal University of Peshawar	Apache	No	No	Apache, Bootstrap

islamiyat.uop.edu.pk	200	Department of Islamiyat	Apache	No	No	Apache, Bootstrap
pharmacy.uop.edu.pk	200	Department of Pharmacy	Apache	No	No	CodeIgniter, Apache, PHP 5.6
www.uop.edu.pk	200	University of Peshawar	Apache	No	No	Apache, PHP 5.6
bg.uop.edu.pk	200	University of Peshawar	Apache	No	No	Apache, PHP 5.6
virtual.uop.edu.pk	200	University of Peshawar	Apache	No	No	Apache, PHP 5.6
uop.edu.pk	200	University of Peshawar	Apache	No	No	Apache, PHP 5.6
pashtoacademy.uop.edu.pk	301 → 301 → 404	404 Not Found	Apache	https://pwr.nu.edu.pk/pacademy/	No (initial URL)	Apache HTTP Server

```

(kali@kali)-[~]
$ httpx -l uopDomains.txt -sc -title -server -location -tech-detect -follow-redirects -o httpx_results.txt

[INF] Current httpx version v1.9.0 (outdated)
[WRN] UI Dashboard is disabled, Use -dashboard option to enable
https://etc.uop.edu.pk [502] [] [502 Bad Gateway] [nginx/1.24.0 (Ubuntu)] [Nginx:1.24.0,Ubuntu]
https://portal.uop.edu.pk [200] [] [PAKTECH Complete ERP Solution] [Microsoft-IIS/10.0] [Bootstrap:3.3.7,IIS:10.0,Microsoft ASP.NET,SweetAlert,Windows Server,jQuery CDN,jQuery:3.1.0]
https://demo.uop.edu.pk [200] [] [University of Peshawar - Leading Public University in Pakistan] [nginx/1.24.0 (Ubuntu)] [Bootstrap,HSTS,Next.js,Node.js,React,Ubuntu,Webpack]
https://learn.uop.edu.pk [303,200] [] [Dashboard (Guest) | Learning Management System] [nginx/1.24.0 (Ubuntu)] [https://learn.uop.edu.pk/my/] [Moodle,Nginx:1.24.0,PHP,RequireJS,Ubuntu]
https://baragali.uop.edu.pk [200] [] [Baragali Summer Camp - UoP] [nginx/1.24.0 (Ubuntu)] [HSTS,Next.js,Nginx:1.24.0,Node.js,React,Ubuntu,Webpack]
https://szic.uop.edu.pk [302,200] [] [Account Suspended] [LiteSpeed] [https://szic.uop.edu.pk/cgi-sys/suspendedpage.cgi] [HTTP/3,LiteSpeed]
http://mail.uop.edu.pk [404] [] [Error 404 (Not Found)] [ghs]
http://ncepc.uop.edu.pk [200] [] [Home] [Apache/2.4.54 (Unix) OpenSSL/1.1.1s PHP/8.0.25 mod_perl/2.0.12 Perl/v5.34.1] [Apache HTTP Server:2.4.54,Joomla,OpenSSL:1.1.1s,PHP:8.0.25,Perl:5.34.1,UNIX,jQuery,mod_perl:2.0.12]
https://mosaic.uop.edu.pk [200] [] [mosaic.uop.edu.pk - Journal, Newsletter, IR] [LiteSpeed] [Elementor:4.1.4,Google Analytics,HTTP/3,Hostinger,LiteSpeed,LiteSpeed Cache,Litespeed Cache,MySQL,PHP:8.2.30,Site Kit:1.182.0,WordPress Block Editor,WordPress Site Editor,WordPress:6.9.5,jQuery,jQuery Migrate:3.4.1]
https://isc.uop.edu.pk [200] [] [International Studies, Baragali - Conference & Summer School] [LiteSpeed] [HTTP/3,Hostinger,Kadence WP Blocks,Kadence WP Kadence:1.4.3,LiteSpeed,LiteSpeed Cache,Litespeed Cache,MySQL,PHP:8.3.30,Site Kit:1.182.0,The

```

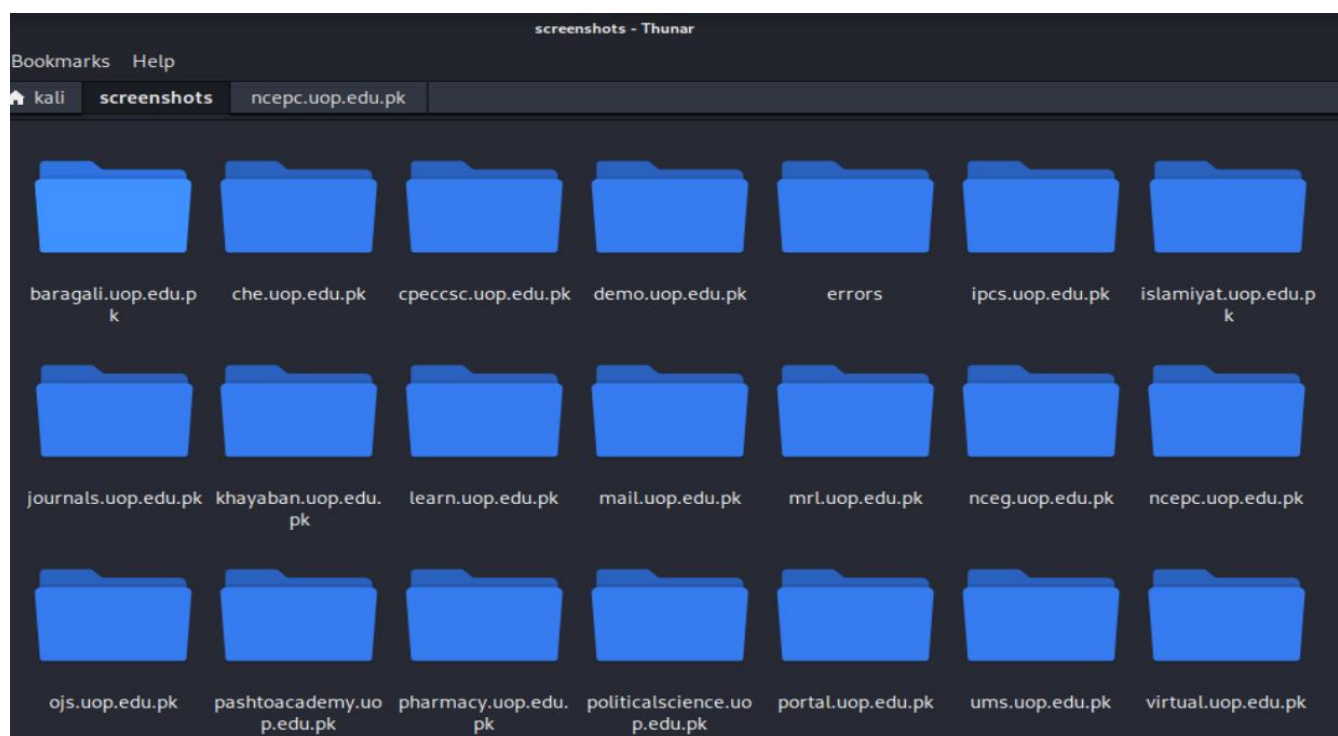
Task 6 — Web Screenshot Collection

Gowitness automates screenshot capture of every live web application discovered in Task 5, providing quick visual triage to spot login pages, admin panels, and exposed dashboards without manually browsing to each host.

6.1 Notable Findings

Type	Subdomain(s) Identified
Login Pages	<i>portal.uop.edu.pk, learn.uop.edu.pk</i>
Admin Panels	<i>portal.uop.edu.pk, ums.uop.edu.pk</i>
API Endpoints	<i>None Identified</i>
Dashboards	<i>learn.uop.edu.pk, ums.uop.edu.pk</i>
Error Pages	<i>etc.uop.edu.pk, mail.uop.edu.pk, szic.uop.edu.pk, pashtoacademy.uop.edu.pk</i>
Development Portals	<i>demo.uop.edu.pk</i>

```
(kali@kali)-[~]
$ gowitness scan file -f live_urls.txt --screenshot-path screenshots
2026/07/23 22:41:46 WARN no writers have been configured. to persist probe results, add writers using --write-* flags
2026/07/23 22:41:50 INFO result 📄 target=https://etc.uop.edu.pk:443 status-code=502 title="502 Bad Gateway" have-screenshot=true
2026/07/23 22:41:51 INFO result 📄 target=https://portal.uop.edu.pk:443 status-code=200 title="PAKTECH Complete ERP Solution" have-screenshot=true
2026/07/23 22:41:51 INFO result 📄 target=https://baragali.uop.edu.pk:443 status-code=200 title="Baragali Summer Camp - UoP" have-screenshot=true
2026/07/23 22:41:51 INFO result 📄 target=https://demo.uop.edu.pk:443 status-code=200 title="University of Peshawar - Leading Public University in Pakistan" have-screenshot=true
2026/07/23 22:41:56 INFO result 📄 target=http://mail.uop.edu.pk:80 status-code=404 title="Error 404 (Not Found)!!1" have-screenshot=true
2026/07/23 22:41:56 INFO result 📄 target=http://ncepc.uop.edu.pk:80 status-code=200 title=Home have-screenshot=true
2026/07/23 22:41:57 INFO result 📄 target=https://learn.uop.edu.pk:443 status-code=200 title="Log in to the site | Learning Management System" have-screenshot=true
2026/07/23 22:41:59 INFO result 📄 target=http://ncepc.uop.edu.pk:443 status-code=400 title="Bad request!" have-screenshot=true
2026/07/23 22:42:11 INFO result 📄 target=https://szic.uop.edu.pk:443 status-code=200 title="Account Suspended" have-screenshot=true
2026/07/23 22:42:12 INFO result 📄 target=https://ums.uop.edu.pk:443 status-code=200 title="Home - UMS" have-screenshot=true
2026/07/23 22:42:18 INFO result 📄 target=http://cpeccsc.uop.edu.pk:80 status-code=200 title="CPEC CSC Conference 2019 | UoP" have-screenshot=true
2026/07/23 22:42:24 INFO result 📄 target=http://khayaban.uop.edu.pk:80 status-code=200 title="Khayaban - خیابان" have-screenshot=true
2026/07/23 22:42:28 INFO result 📄 target=https://politicalscience.uop.edu.pk:443 status-code=200 title="Home - Politics" have-screenshot=true
2026/07/23 22:42:32 INFO result 📄 target=http://mrl.uop.edu.pk:80 status-code=403 title="Apache HTTP Server Test Page powered by CentOS" have-screenshot=true
2026/07/23 22:42:56 INFO result 📄 target=https://mosaic.uop.edu.pk:443 status-code=200 title="" have-screenshot=false
2026/07/23 22:42:59 INFO result 📄 target=https://isc.uop.edu.pk:443 status-code=200 title="" have-screenshot=false
```



Task 7 — Technology Fingerprinting

For every live asset identified, this task documents the underlying technology stack, which informs both the attack surface diagram (Task 8) and the risk assessment (Task 9).

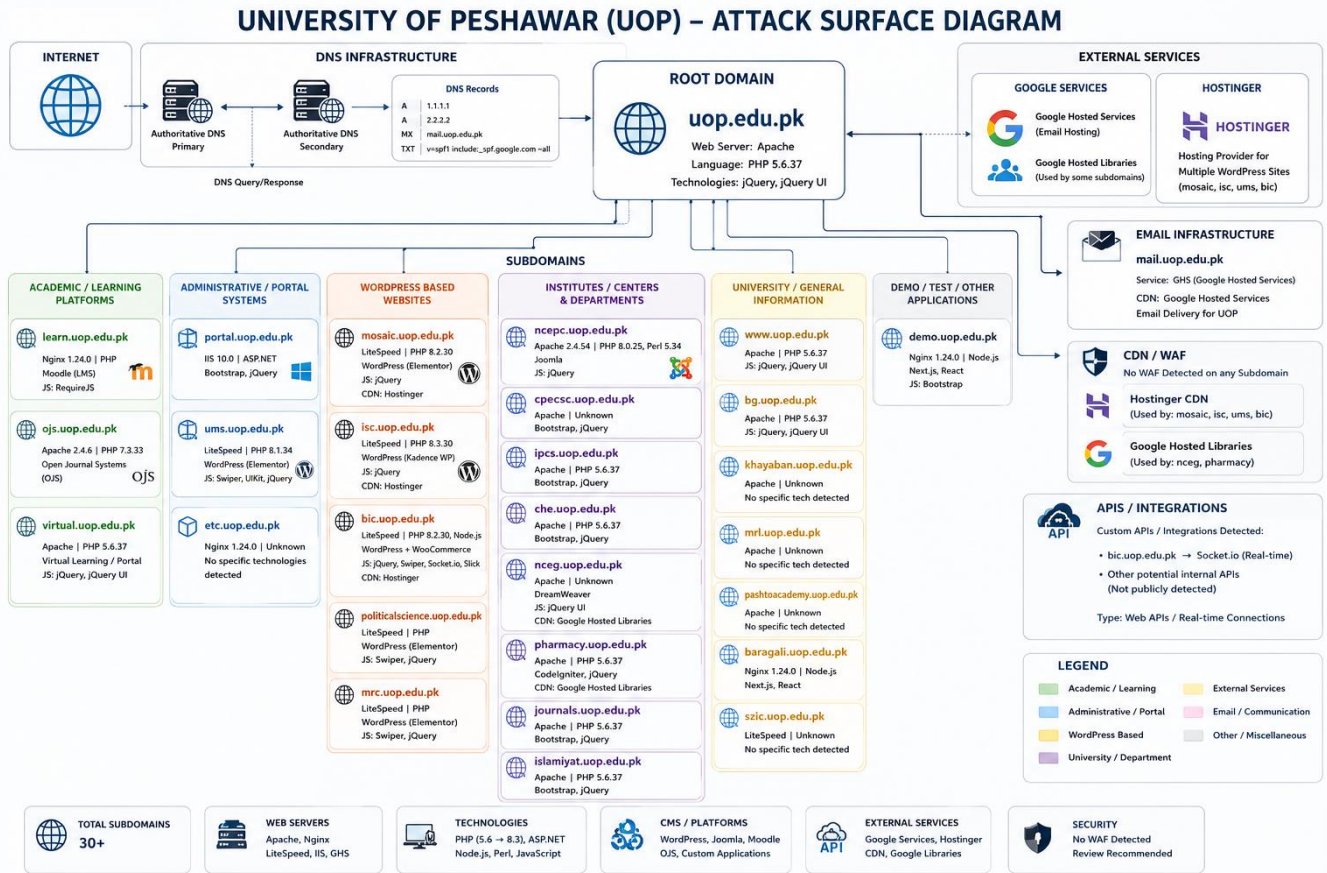
Subdomain	Web Server	Programming Language	Framework	CMS	JavaScript Libraries	CDN	WAF
<i>portal.uop.edu.pk</i>	<i>Microsoft IIS 10.0</i>	<i>ASP.NET</i>	<i>Bootstrap</i>	<i>None</i>	<i>jQuery</i>	<i>None Detected</i>	<i>None Detected</i>
<i>demo.uop.edu.pk</i>	<i>Nginx 1.24.0</i>	<i>Node.js</i>	<i>Next.js, React</i>	<i>None</i>	<i>Bootstrap</i>	<i>None Detected</i>	<i>None Detected</i>
<i>learn.uop.edu.pk</i>	<i>Nginx 1.24.0</i>	<i>PHP</i>	<i>Moodle</i>	<i>Moodle</i>	<i>RequireJS</i>	<i>None Detected</i>	<i>None Detected</i>
<i>baragali.uop.edu.pk</i>	<i>Nginx 1.24.0</i>	<i>Node.js</i>	<i>Next.js, React</i>	<i>None</i>	<i>None Detected</i>	<i>None Detected</i>	<i>None Detected</i>
<i>etc.uop.edu.pk</i>	<i>Nginx 1.24.0</i>	<i>Unknown</i>	<i>None</i>	<i>None</i>	<i>None</i>	<i>None Detected</i>	<i>None Detected</i>

<i>szic.uop.edu.pk</i>	<i>LiteSpeed</i>	<i>Unknown</i>	<i>None</i>	<i>None</i>	<i>None</i>	<i>None Detected</i>	<i>None Detected</i>
<i>mail.uop.edu.pk</i>	<i>GHS</i>	<i>Unknown</i>	<i>None</i>	<i>None</i>	<i>None</i>	<i>Google Hosted Services</i>	<i>None Detected</i>
<i>ncepc.uop.edu.pk</i>	<i>Apache 2.4.54</i>	<i>PHP 8.0.25, Perl 5.34</i>	<i>Joomla</i>	<i>Joomla</i>	<i>jQuery</i>	<i>None Detected</i>	<i>None Detected</i>
<i>mosaic.uop.edu.pk</i>	<i>LiteSpeed</i>	<i>PHP 8.2.30</i>	<i>Elementor</i>	<i>WordPress</i>	<i>jQuery</i>	<i>Hostinger</i>	<i>None Detected</i>
<i>isc.uop.edu.pk</i>	<i>LiteSpeed</i>	<i>PHP 8.3.30</i>	<i>Kadence WP</i>	<i>WordPress</i>	<i>jQuery</i>	<i>Hostinger</i>	<i>None Detected</i>
<i>ums.uop.edu.pk</i>	<i>LiteSpeed</i>	<i>PHP 8.1.34</i>	<i>Elementor</i>	<i>WordPress</i>	<i>Swiper, UIKit, jQuery</i>	<i>Hostinger</i>	<i>None Detected</i>
<i>bic.uop.edu.pk</i>	<i>LiteSpeed</i>	<i>PHP 8.2.30, Node.js</i>	<i>Elementor</i>	<i>WordPress + WooCommerce</i>	<i>jQuery, Swiper, Socket.io, Slick</i>	<i>Hostinger</i>	<i>None Detected</i>
<i>politicalscience.uop.edu.pk</i>	<i>LiteSpeed</i>	<i>PHP</i>	<i>Elementor</i>	<i>WordPress</i>	<i>Swiper, jQuery</i>	<i>None Detected</i>	<i>None Detected</i>
<i>cpeccsc.uop.edu.pk</i>	<i>Apache</i>	<i>Unknown</i>	<i>Bootstrap</i>	<i>None</i>	<i>jQuery</i>	<i>None Detected</i>	<i>None Detected</i>
<i>khayaban.uop.edu.pk</i>	<i>Apache</i>	<i>Unknown</i>	<i>None</i>	<i>None</i>	<i>None</i>	<i>None Detected</i>	<i>None Detected</i>
<i>mrl.uop.edu.pk</i>	<i>Apache</i>	<i>Unknown</i>	<i>None</i>	<i>None</i>	<i>None</i>	<i>None Detected</i>	<i>None Detected</i>
<i>ojs.uop.edu.pk</i>	<i>Apache 2.4.6</i>	<i>PHP 7.3.33</i>	<i>None</i>	<i>Open Journal Systems</i>	<i>None Detected</i>	<i>None Detected</i>	<i>None Detected</i>
<i>mrc.uop.edu.pk</i>	<i>LiteSpeed</i>	<i>PHP</i>	<i>Elementor</i>	<i>WordPress</i>	<i>Swiper, jQuery</i>	<i>None Detected</i>	<i>None Detected</i>
<i>ipcs.uop.edu.pk</i>	<i>Apache</i>	<i>PHP 5.6.37</i>	<i>Bootstrap</i>	<i>None</i>	<i>jQuery</i>	<i>None Detected</i>	<i>None Detected</i>

<i>che.uop.edu.pk</i>	<i>Apache</i>	<i>PHP 5.6.37</i>	<i>Bootstrap</i>	<i>None</i>	<i>jQuery</i>	<i>None Detect ed</i>	<i>None Detect ed</i>
<i>nceg.uop.edu.pk</i>	<i>Apache</i>	<i>Unknown</i>	<i>DreamWeaver</i>	<i>None</i>	<i>jQuery UI</i>	<i>Google Hosted Libraries</i>	<i>None Detect ed</i>
<i>journals.uop.edu.pk</i>	<i>Apache</i>	<i>PHP 5.6.37</i>	<i>Bootstrap</i>	<i>None</i>	<i>jQuery</i>	<i>None Detect ed</i>	<i>None Detect ed</i>
<i>islamiyat.uop.edu.pk</i>	<i>Apache</i>	<i>PHP 5.6.37</i>	<i>Bootstrap</i>	<i>None</i>	<i>jQuery</i>	<i>None Detect ed</i>	<i>None Detect ed</i>
<i>pharmacy.uop.edu.pk</i>	<i>Apache</i>	<i>PHP 5.6.37</i>	<i>CodeIgniter</i>	<i>None</i>	<i>jQuery</i>	<i>Google Hosted Libraries</i>	<i>None Detect ed</i>
<i><u>www.uop.edu.pk</u></i>	<i>Apache</i>	<i>PHP 5.6.37</i>	<i>None</i>	<i>None</i>	<i>jQuery, jQuery UI</i>	<i>None Detect ed</i>	<i>None Detect ed</i>
<i>bg.uop.edu.pk</i>	<i>Apache</i>	<i>PHP 5.6.37</i>	<i>None</i>	<i>None</i>	<i>jQuery, jQuery UI</i>	<i>None Detect ed</i>	<i>None Detect ed</i>
<i>virtual.uop.edu.pk</i>	<i>Apache</i>	<i>PHP 5.6.37</i>	<i>None</i>	<i>None</i>	<i>jQuery, jQuery UI</i>	<i>None Detect ed</i>	<i>None Detect ed</i>
<i>uop.edu.pk</i>	<i>Apache</i>	<i>PHP 5.6.37</i>	<i>None</i>	<i>None</i>	<i>jQuery, jQuery UI</i>	<i>None Detect ed</i>	<i>None Detect ed</i>
<i>pashtoacademy.uop.edu.pk</i>	<i>Apache</i>	<i>Unknown</i>	<i>None</i>	<i>None</i>	<i>None</i>	<i>None Detect ed</i>	<i>None Detect ed</i>

Task 8 — Attack Surface Mapping

A visual attack surface diagram consolidates all reconnaissance findings the root domain, subdomains, DNS relationships, external services, email infrastructure, web applications, APIs, and any CDN/WAF into a single reference diagram for stakeholders.



Task 9 — Risk Identification

Based only on reconnaissance findings — with no exploitation performed — at least 10 potential security risks must be identified, explained, and paired with a mitigation recommendation.

#	Finding	Description	Potential Impact	Recommendation
1	Exposed Administrative Portal	<i>portal.uop.edu.pk</i> appears to be an administrative portal running Microsoft IIS and ASP.NET.	May become a target for brute-force attacks, credential theft, or unauthorized access attempts.	Protect with Multi-Factor Authentication (MFA), strong passwords, IP restrictions, and continuous monitoring.
2	Public Development Environment	<i>demo.uop.edu.pk</i> appears to be a development/demo environment using Node.js and Next.js.	Development systems may expose unfinished features, debugging information, or configuration details.	Restrict access to authorized users or deploy behind authentication/VPN.
3	Large Attack Surface	More than 25 publicly accessible subdomains were discovered.	A larger attack surface increases the likelihood of forgotten or vulnerable services.	Regularly review and remove unused subdomains and maintain an up-to-date asset inventory.
4	Multiple CMS Platforms	WordPress, Joomla, Moodle, and Open Journal Systems (OJS) are deployed across different subdomains.	Multiple CMS platforms require separate maintenance and increase the number of potential attack vectors.	Keep all CMS installations, themes, and plugins updated and remove unused components.
5	Outdated PHP Versions	Several websites are running PHP 5.6.37 and 7.3.33, which have reached End-of-Life (EOL).	Unsupported PHP versions no longer receive security updates, increasing the risk of known vulnerabilities.	Upgrade all systems to a supported PHP 8.x release.
6	Older Apache Version	<i>ojs.uop.edu.pk</i> is running Apache 2.4.6, an older server version.	Older software may contain publicly known vulnerabilities if security patches are not applied.	Upgrade Apache to a supported version or ensure all security patches are installed.
7	No Web Application Firewall (WAF) Detected	No WAF was identified for the discovered web applications.	Applications may have less protection against common web attacks such as SQL injection, XSS,	Deploy a Web Application Firewall such as Cloudflare WAF, ModSecurity, or Azure WAF.

			<i>and automated scanning.</i>	
8	<i>Technology Stack Disclosure</i>	<i>Server software, PHP versions, frameworks, CMSs, and JavaScript libraries were identifiable through reconnaissance.</i>	<i>Technology fingerprinting helps attackers identify technologies and target known vulnerabilities.</i>	<i>Minimize unnecessary version disclosure through server configuration and security headers.</i>
9	<i>Public Email Infrastructure</i>	<i>mail.uop.edu.pk is publicly accessible and hosted using Google Hosted Services.</i>	<i>Email services are common targets for phishing, password spraying, and credential attacks.</i>	<i>Enforce MFA, configure SPF, DKIM, and DMARC, and continuously monitor login activity.</i>
10	<i>Potential API Exposure</i>	<i>bic.uop.edu.pk uses Socket.io, indicating the presence of real-time communication/API endpoints.</i>	<i>Public APIs may expose sensitive functionality if authentication or authorization is weak.</i>	<i>Secure APIs with authentication, authorization, rate limiting, and input validation.</i>

Task 10 — Reconnaissance Automation

To streamline repetitive reconnaissance steps, a script automating at least three of the following tasks must be created: WHOIS lookup, DNS lookup, subdomain enumeration, live host validation, saving results to files, and organizing output into folders.

Script

```

kali@kali: ~
GNU nano 8.7.1 recon.sh
#!/bin/bash

# Ask user for target domain
echo "Enter target domain :"
read DOMAIN

# Create output directory
mkdir -p Recon_${DOMAIN}

echo "Output folder created."

# whois details
echo "Performing WHOIS lookup ..."
whois $DOMAIN > Recon_${DOMAIN}/whois.txt

# DNS Lookup
echo "Performing DNS lookup ..."
dig ANY $DOMAIN > Recon_${DOMAIN}/dns.txt

# Subdomain Enumeration using subfinder
echo "Enumerating subdomains ..."
subfinder -d $DOMAIN -silent > Recon_${DOMAIN}/subdomains.txt

# Live Host Validation using httpx
echo "Checking live hosts ..."
httpx -l Recon_${DOMAIN}/subdomains.txt \
-silent \
-title \
-tech-detect \
> Recon_${DOMAIN}/live_hosts.txt

# finished
echo ""
echo "===== "
echo " Reconnaissance Completed Successfully!"
echo " Results saved in: Recon_${DOMAIN}/"
echo "===== "

```

```

kali@kali: ~
(kali@kali)-[~]
$ ./recon.sh
Enter target domain :
uetmardan.edu.pk
Output folder created.
Performing WHOIS lookup ...
Performing DNS lookup ...
Enumerating subdomains ...
Checking live hosts ...

=====
Reconnaissance Completed Successfully!
Results saved in: Recon_uetmardan.edu.pk/
=====

(kali@kali)-[~]
$ tree Recon_uetmardan.edu.pk
Recon_uetmardan.edu.pk
├── dns.txt
├── live_hosts.txt
├── subdomains.txt
└── whois.txt

1 directory, 4 files

(kali@kali)-[~]
$ ls Recon_uetmardan.edu.pk
dns.txt live_hosts.txt subdomains.txt whois.txt

(kali@kali)-[~]
$

```